

# CONFERENCE PROCEEDINGS

## VIETNAM AI SUMMIT 2024

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## Paper #1: Optimizing LLMs for Vietnamese Language

*Author: Isabella Martinez - NUS Singapore*

[DOI: 10.1234/icai.2026.101](https://doi.org/10.1234/icai.2026.101)

### **Abstract:**

No abstract provided.

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## Paper #2: Attention Is All You Need

*Author: Christopher Hernandez - ETH Zurich*

### **Abstract:**

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

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## Paper #3: Generative Adversarial Nets

*Author: Daniel Rodriguez - University of Tokyo*

### **Abstract:**

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model  $G$  that captures the data distribution, and a discriminative model  $D$  that estimates the probability that a sample came from the training data rather than  $G$ . The training procedure for  $G$  is to maximize the probability of  $D$  making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions  $G$  and  $D$ , a unique solution exists, with  $G$  recovering the training data distribution and  $D$  equal to  $1/2$  everywhere. In the case where  $G$  and  $D$  are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

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## Paper #4: Scalable Vision Transformers with Hierarchical Attention

*Author: Tran Reviewer - FPT Software*

**Abstract:**

The recently proposed Visual image Transformers (ViT) with pure attention have achieved promising performance on image recognition tasks, such as image classification. However, the routine of the current ViT model is to maintain a full-length patch sequence during inference, which is redundant and lacks hierarchical representation. To this end, we propose a Hierarchical Visual Transformer (HVT) which progressively pools visual tokens to shrink the sequence length and hence reduces the computational cost, analogous to the feature maps downsampling in Convolutional Neural Networks (CNNs). It brings a great benefit that we can increase the model capacity by scaling dimensions of depth/width/resolution/patch size without introducing extra computational complexity due to the reduced sequence length. Moreover, we empirically find that the average pooled visual tokens contain more discriminative information than the single class token. To demonstrate the improved scalability of our HVT, we conduct extensive experiments on the image classification task. With comparable FLOPs, our HVT outperforms the competitive baselines on ImageNet and CIFAR-100 datasets

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## **Paper #5: Denoising Diffusion Probabilistic Models for Image Synthesis**

*Author: Daniel Rodriguez - University of Tokyo*

**Abstract:**

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at

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## **Paper #6: Human-Level Reasoning: A Comparative Study of LLMs**

*Author: Le Author - Hanoi University*

**Abstract:**

Evaluating reasoning ability in Large Language Models (LLMs) is important for advancing artificial intelligence, as it transcends mere linguistic task performance. It involves understanding whether these models truly understand information, perform inferences, and are able to draw conclusions in a logical and valid way. This study compares logical and abstract reasoning skills of several LLMs - including GPT, Claude, DeepSeek, Gemini, Grok, Llama, Mistral, Perplexity, and Sabiá - using a set of eight custom-designed reasoning questions. The LLM results are benchmarked against human performance on the same tasks, revealing significant differences and indicating areas where LLMs struggle with deduction.

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